## • Medical Science, Anthropology, Zoology

**64. With reference to the recent developments in science, which one of the following statements is not correct?**

(a) Functional chromosomes can be created by joining segments of DNA taken from cells of different species.
(b) Pieces of artificial functional DNA can be created in laboratories.
(c) A piece of DNA taken out from an animal cell can be made to replicate outside a living cell in a laboratory.
(d) Cells taken out from plants and animals can be made to undergo cell division in laboratory petri dishes.

**Ans : (a)** Biotechnology is the development of various stuff or services through the actions of material by adopting scientific and engineering principles. Cloning DNA fingerprinting, biochips etc are studied under biotechnology. Functional DNA, rather than functional chromosomes, are produced by combining DNA segments taken from cells of different races. Hence statement (a) is false.
The application of biotechnology can also be seen as follows-
(1) Artificial Functional DNA parts can be created in laboratories using this technology.
(2) Under this any parts of the DNA extracted from an animal cell can be replicated outside the living cell in the laboratory.
(3) Cell extracted from plants and animals may undergo cell division in laboratory petri dish.
Based on the above interpretation option B,C and D are correct.

**65. In the context of wearable technology, which of the following tasks is/are accomplished by wearable devices?**

**1. Location identification of a person**
**2. Sleep monitoring of a person**
**3. Assisting the hearing impaired person**

**Select the correct answer using the code given below.**

(a) 1 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

**Ans : (d)** Wearable technology is a technology that has the ability to connect to the internet. Wearable devices are implanted into clothing through fashion technology. Information about the users' location, sleep monitoring, heart rate, blood pressure, time spent in exercise, biochemical emissions, hearing defects etc, can be obtained through their device. Presently, this smart technology, mobile application, computing has become very popular due to digital watch and broadband connectivity. Devices used for hearing impaired people are good examples of this.

**66. 'RNA interference (RNA)' technology has gained popularity in the last few years. Why?**

**1. It is used in developing gene silencing therapies.**
**2. It can be used in developing therapies for the treatment of cancer.**
**3. It can be used to develop hormone replacement therapies**
**4. It can be used to produce crop plants that are resistant to viral pathogens.**

**Select the correct answer using the code given below :**

(a) 1, 2 and 4 (b) 2 and 3
(c) 1 and 3 (d) 1 and 4 only

**Ans : (a)** RNA is a gene- silencing technology that inhibits protein synthesis in target cells using double-stranded RNA.
RNA has huge significance within the Indian context, considering the deep-seated resistance over the years to BT cotton and other genetically modified seeds. RNA-reliant solutions could be a viable alternative. RNA is used in functional genomics (systematic analysis of loss of function phenotypes induced by RNA triggers) and developing therapies for the treatment of viral infection dominant disorders, neurological disorders, and many types of cancers (in vivo inactivation of gene products linked to human disease progression and pathology).

**67. Which of the following are the reasons for the occurrence of multi-drug resistance in-microbial pathogens in India?**

**(1) Genetic predisposition of some people.**
**(2) Taking incorrect doses of antibiotics to cure diseases.**
**(3) Using antibioties in livestock farming.**
**(4) Multiple chronic diseases in some people.**

**Select the correct answer using the codes given below :**

(a) 1 and 2 (b) 2 and 3 only
(c) 1, 3 and 4 (d) 2, 3 and 4

**Ans : (b)** The following are the major reasons for multi-drug resistance in Indian microbial diseases-improper use of antibiotics for the treatment of diseases, improper use of antibiotics in livestock farming and apart from these in the treatment of any type of disease, not taking entire course of antibiotics as well as the use of the wrong medicines.

**68. What is Cas9 protein that is often mentioned in news?**

(a) A molecular scissors used in targeted gene editing
(b) A biosensor used in the accurate detection of pathogens in patients
(c) A gene that makes plants pest-resistant
(d) A herbicidal substance synthesized in genetically modified crops

**Ans : (a)** The key step in editing an organism's genome is selective targeting of a specific sequence of DNA. Two biological macromolecules, the Cas9 protein and guide RNA; interact to form a complex that can identify target sequences with high selectivity. The Cas-9 protein is responsible for locating and cleaving target DNA both in natural and in artificial CRISPR/Cas9 systems.



**69. Which one of the following statements is not correct?**

(a) Hepatitis B virus is transmitted much like HIV.
(b) Hepatitis B, unlike Hepatitis C, does not have a vaccine.
(c) Globally, the number of people infected with Hepatitis B and C viruses are several times more than those infected with HIV.
(d) Some of those infected with Hepatitis B and C viruses do not show the symptoms for many years.

**Ans : (b)** Hepatitis B is a common infectious disease in the liver. The disease is caused by the 'Hepatitis B' virus. This disease is spread by blood, unprotected sex, infected needles, etc. This infection happens unknowingly; that is why it is also called a silent infection. There is a high possibility of having liver cirrhosis due to this infection. According to the assessment of the World Health Organization, about 240 million people have Hepatitis-B globally and about 71 million people with the hepatitis-C virus. Several vaccines for the prevention of Hepatitis B virus infection were developed by Maurice Hilleman. Currently, vaccines are being prepared with the help of recombinant DNA technology for the prevention of this virus.

## • Agriculture, Animal Husbandry & Dairy

**70. Consider the following statements:**

**1. Agricultural soils release nitrogen oxides into environment.**
**2. Cattle release ammonia into environment.**
**3. Poultry industry releases reactive nitrogen compounds into environment.**

**Which of the statements given above is/are correct?**

(a) 1 and 3 only (b) 2 and 3 only
(c) 2 only (d) 1, 2 and 3

**Ans : (d)** Agricultural soils release nitrogen oxides into the environment. According to an estimate since 2010 agricultural soils in India contribute to the emission of more than 70 percent nitrogen oxides in the environment. Large quantities of ammonia are released from the cattle in the form of ammonia gas. It is estimated that 80 percent of the ammonia emission in the environment are from cattle, which is increasing at the rate of 1 percent per year. The poultry industry releases large quantities of reactive nitrogen compounds, which was estimated to be 0.415 tonnes in 2016 and by 2030 is estimated to be 1.089 tonnes.

## • Science & Technology

**71. Consider the following statements :**

**1. According to the Indian Patents Act, a biological Process to create a seed can be patented in India.**
**2. In India, there is no Intellectual Property Appellate Board.**
**3. Plant varieties are not eligible to be patented in India.**

**Which of the statements given above is/are correct?**

(a) 1 and 3 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

**Ans : (c)** Article 3 (j) of India's patent act. excludes from patentability of "plants and animals in whole or in any part, thereof other than microorganisms but including seeds, varieties and species, and essentially biological processes for production or propagation of plants and animals" Hence statement 1 is not correct. The Intellectual Property Appellate Board (IPAB) was constituted on 15 September, 2003 by the Indian Government to hear and resolve the appeals against the decisions of the registrar under the Indian Government to hear and resolve the appeals against the decisions of the registrar under the Indian Trademarks Act, 1999 and the Geographical Indication of Goods (Registration and Protection Act, 1999.) Hence, statement 2 is not correct.

**72. Why are dewdrops not formed on a cloudy night?**

(a) Clouds absorb the radiation released from the Earth's surface.
(b) Clouds reflect back the Earth's radiation.
(c) The Earth's surface would have low temperature on cloudy nights.
(d) Clouds deflect the blowing wind to ground level.

**Ans : (b)** Earth's temperature is high during the day; consequently water evaporates and at night there is radiation of heat from the earth and if the sky is clean then it is deposited in the radiated heat environment and the earth becomes cold. Due to the low temperature of the earth, condensation of water vapour in the atmosphere occurs, and as a result it is produced in the form of dew drops. When there are clouds in the sky, they again reflect the heat radiated by the earth back to the earth. As a result, the temperature of the earth is not reduced enough to condense the atmosphere. As a result dew droplets are not formed.

**73. For the measurement/estimation of which of the following are satellite images/remote sensing data used?**

**1. Chlorophyll content in the vegetation of a specific location.**
**2. Greenhouse gas emissions from rice paddies of a specific location.**
**3. Land surface temperatures of a specific location.**

**Select the correct answer using the code given below.**

(a) 1 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

**Ans : (d)** Through different vegetation index (NDVI) and Near-infrared radiation, remote sensing satellites can detect healthy vegetation in agriculture. Healthy vegetation reflects green light and absorbs red and blue light. The green light that our eyes see in chlorophyll created by plants during photosynthesis. Chlorophyll will reflect more light in the green and near- infrared spectrum compared to other wavelengths.
Remote sensing satellite data offers the possibility for measuring land surface temperature over the entire earth